A Photovoltaic Hot-Spot Fault Detection Network for Aerial Images Based on Progressive Transfer Learning and Multiscale Feature Fusion

Shuai Hao, Jiahao Li, Xu Ma, Siya Sun, Zhuo Tian, Tianqi Li, and Yifeng Hou

***Abstract*— The number of samples is one of the key factors affecting the performance of deep learning-based detection networks. Aiming at the problem that the detection network is difficult to accurately detect the hot-spot fault targets under the condition of small samples, a photovoltaic hot-spot fault detection network based on progressive transfer learning and multiscale feature fusion is proposed. First, a large number of artificial hot-spot samples are generated through the artificial model, and the mixed dataset containing real and artificial samples is constructed to improve the data diversity. On this basis, a pre-trained model based on artificial samples is established to learn the shallow features of hot-spot faults. Then, to fuse the multiscale features and improve feature aggregation ability of detection network, a novel feature pyramid structure based on reparameterized generalized and multiscale feature fusion (RepG-MSFF) is designed. Moreover, to balance the detection accuracy and speed, the spatial and channel reconstruction convolution (SCConv) is utilized to replace conventional convolution in the backbone network. Finally, to further accurately locate hot-spot targets, an adaptive threshold focal loss (TFL) function is introduced. The experimental results indicate that, in three different scenarios datasets, the detection accuracy can reach 87.9%, 88.6%, and 87.7%, respectively, which is higher than that of other nine detection algorithms.**

***Index Terms*— Artificial sample, hot-spot detection, multiscale feature fusion, parallel vision, progressive transfer learning.**

I. Introduction

With the high attention of countries to energy security and environmental protection, photovoltaic power generation has been widely used around the word, which can promote low-carbon economic transformation and sustainable development [1]. However, with the rapid expansion of photovoltaic industry, the security challenges faced by photovoltaic power generation systems are becoming increasingly severe. Due to long-term exposure to outdoor environments, photovoltaic modules are easy to be blocked by external foreign objects, resulting in hot-spot effect [2], [3]. The existence of hot-spot effect could shorten the lifespan of photovoltaic module, and in serious cases will lead to fire accidents. Therefore, it is necessary to regularly and accurately detect hot-spot faults.

The traditional hot-spot fault detection methods mainly rely on manual inspection, which needs operation and maintenance personnel to hold infrared thermal imaging equipment for inspection [4], [5]. This method not only requires a lot of manpower and material resources, but also requires high professional knowledge and skills of inspection personnel. As the scale of photovoltaic power stations continues to expand, the traditional manual detection methods have been difficult to meet the needs of daily maintenance [6]. With the rapid development of computer vision, intelligent inspections have gradually replaced manual detection [7]. It uses unmanned aerial vehicles (UAVs) to carry infrared thermal imaging equipment and accurately locates the hot-spot faults through object detection algorithms. The application of intelligent inspection technology greatly improves the efficiency of daily maintenance [8].

In order to effectively detect hot-spot faults, a lot of research has been carried out. Wu and Hao [9] introduced the attention mechanism and feature pyramid structure on the basis of faster-RCNN, which effectively improves the detection accuracy for small targets; however, the introduction of attention modules increases the computational complexity and reduces inference speed. Lei et al. [10] designed a hot-spot detection method that combines segmentation and detection, which uses Deeplabv3+ model for the fine segmentation of photovoltaic panels, so as to reduce the interference of complex background on hot-spot targets detection; however, this method is still difficult to identify multiscale hot-spot targets. Zhao et al. [11] designed an SNCF-Net for hot-spot detection, which enhances the utilization of fault features by using the neighborhood correlation feature modules and adaptive scale-aware attention mechanism; however, in this method, small hot-spot targets are easily submerged in complex backgrounds, thereby reducing detection accuracy. Sun et al. [12] proposed a hot-spot detection method based on YOLOv5, which can realize the detection of small targets; however, this method requires a lot of computing resources and is difficult to extract multiscale hot-spot features. In practical scenarios, hot-spot targets often exhibit multiscale features, which make it important to improve the multiscale feature fusion capability of detection model. By effectively integrating

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The authors are with the College of Electrical and Control Engineering, Xi'an University of Science and Technology, Xi'an 710054, China (e-mail: haoxust@163.com; lijhxa@163.com; maxu@xust.edu.cn; Sunsiya412@xust.edu.cn; tianzhuoxust@163.com; altq8792@163.com; 1149989334@qq.com).

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multiscale information, the model can more accurately identify and locate these multiscale targets. Li et al. [13] proposed a multiscale factor joint learning for hyperspectral image, which establishes multiscale information interaction units through a direction-aware spatial context aggregation module, effectively improving the processing efficiency of different scale factors. Wang et al. [14] designed a neural architecture search feature pyramid network, which utilizes neural architecture search technology to effectively capture target features of different scales.

In addition, the above deep learning-based hot-spot detection algorithms are driven by mass data, and the detection performance depends more on the quantity and quality of training samples. However, in the daily inspection of photovoltaic power stations, photovoltaic modules are in normal working state most of the time, so fault images are very scarce. Therefore, the number of training samples and the multiscale characteristics of hot-spot faults become important factors limiting the performance of detection network. To address this issue, in the framework of parallel vision, a novel method based on progressive transfer learning and multiscale feature fusion is proposed. The main innovations are as follows.

1) To solve the problem of a few common features in cross-domain transfer learning, a progressive transfer learning method is proposed. In sample expansion, a large number of artificial hot-spot samples are generated through the artificial model, and the mixed dataset containing real and artificial samples is constructed to improve the data diversity. On this basis, a pre-trained model based on artificial samples is established to learn the shallow features of hot-spot faults.
2) To efficiently fuse multiscale features and improve the feature aggregation capacity of network, a reparameterized generalized and multiscale feature fusion (RepG-MSFF) structure is designed in the neckwork, which can capture the high-level semantic features and low-level spatial features of hot-spot faults.
3) To balance the detection accuracy and inference speed, the convolutional module in original network is replaced by the spatial and channel reconstruction convolution (SCConv) in the backbone network, which can achieve the compression optimization of model structure.
4) To further accurately locate hot-spot targets, an adaptive threshold focal loss (TFL) function is introduced to separate hot-spot faults from complex backgrounds, which can improve the focus of detection model on hot-spot targets.

II. Proposed Methods

A. Hot-Spot Fault Detection Method Based on Progressive Transfer Learning

In cross-domain transfer learning, when there are a few common features between the source and target domain, it is easy to cause ineffective transfer or even negative transfer. Therefore, to effectively solve the issue of limited common features in cross-domain transfer learning, in the framework of parallel vision, this article generates a lot of photovoltaic hot-spot fault datasets that are similar to real samples features through artificial modeling, and takes these datasets as source domain training samples. These artificial samples are highly similar to the real fault samples, which effectively reduces the feature difference between source and target domain, so that the pre-trained model on the source domain can extract the key features consistent with the real sample. By training on rich artificial datasets, a pre-trained model is obtained. Then through the transfer learning mechanism based on model parameters, the effective transfer of network model parameters is realized.

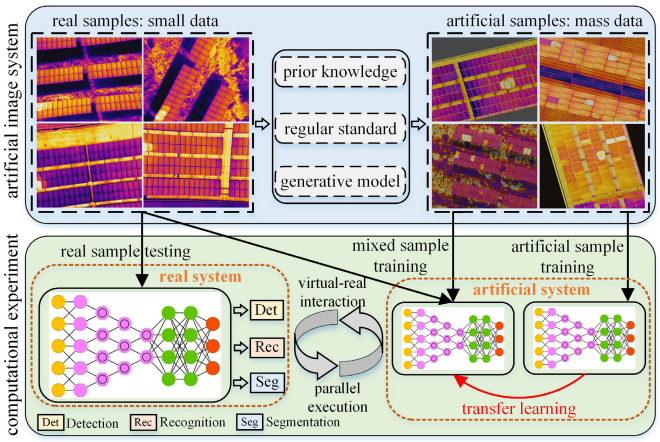


Fig. 1. Basic framework of parallel vision.

1) Basic Framework of Parallel Vision: Parallel vision is an intelligent vision computing method based on artificial system, computational experiment, and parallel execution theory [15], [16], as shown in Fig. 1.

In the framework of parallel vision, the hot-spot fault detection is divided into the following three steps.

1) Artificial image system. By simulating the geometric structure, fault characteristics, and possible environmental conditions of real hot-spot fault samples, a large-scale and diverse dataset of artificial samples are generated.
2) Computational experiment. Due to the differences in distribution between artificial and real samples from different fields, the model learned from artificial samples cannot be directly used for real samples detection. In order to ensure domain adaptability, the pre-trained model is obtained by using the artificial samples of hot-spot faults, which can enable detection model to learn the shallow features of hot-spot faults first. Then, the mixed dataset containing real and artificial samples is constructed, and a pre-trained model is loaded to jointly train, which can help the model accelerate convergence and improve its generalization ability. In the selection of training dataset, the appropriate proportion of artificial and real samples is selected to generate a more robust target detector.
3) Parallel execution. On the basis of computational experiment, the detection results of the artificial system are compared with those of the real system through virtual–real interaction and parallel execution, so as

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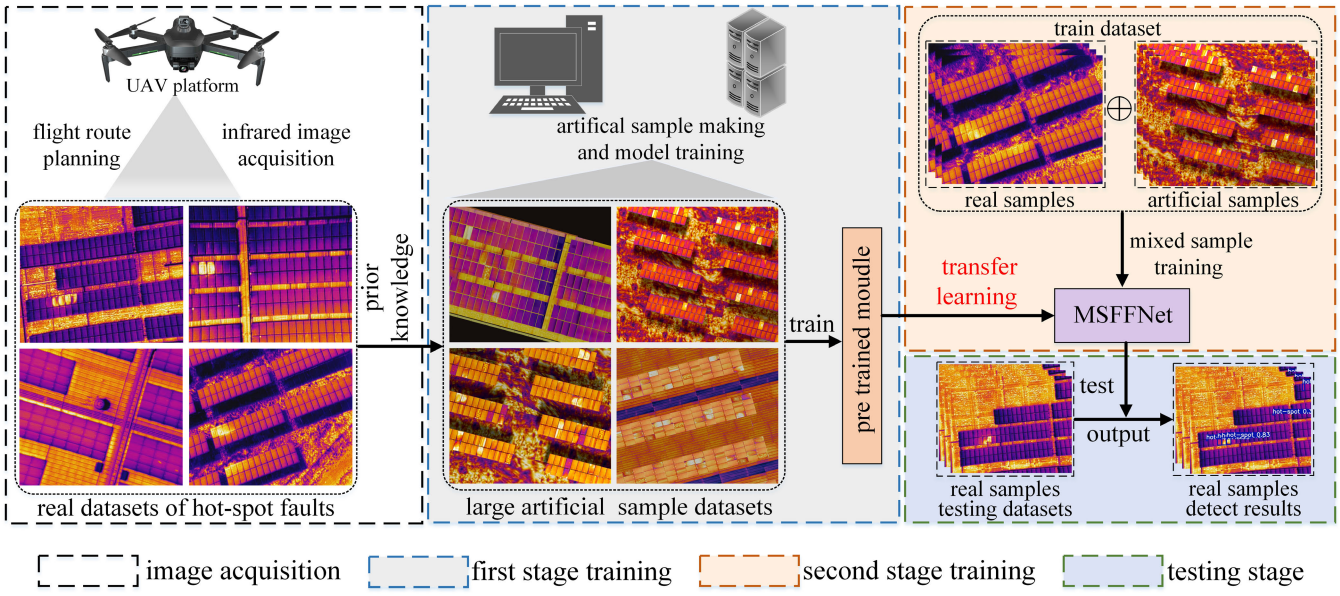


Fig. 2. Detection model based on progressive transfer learning.

TABLE I
APPLICATION SCENARIOS OF TRANSFER LEARNING METHODS

Transfer method	Application scenario
model parameter	the source data is same or similar to target data
feature extraction	the source data is not significant differences to target data
feature mapping	the source data is significant differences to target data

to constantly adjust and optimize the detection model, ensuring its stronger robustness and generalization ability.

2) Progressive Transfer Learning Based on Mixed Samples: Currently, to achieve a better detection performance, the target detection model requires sufficient samples for training [17]. Therefore, to solve the problem of weak detection ability and low robustness in the case of small samples, a training method based on progressive transfer learning is proposed, as shown in Fig. 2. In first stage training, a lot of artificial samples are utilized to train a pre-trained model, which can learn the shallow features of hot-spot faults. In second stage training, the pre-trained model is loaded with mixed samples for training, which makes detection model focus on deep features of hot-spot faults. In testing stage, the real samples of hot-spot faults are used for testing.

In the training method based on transfer learning, it includes three methods: model parameter, feature extraction, and feature mapping [18], [19], as shown in Table I. Due to the similarity between artificial hot-spot fault samples and real samples, the transfer method based on model parameter is adopted in this article.

3) Generation of Artificial Samples for Hot-Spot Faults: To improve the sample diversity, sample augment is the most direct and effective method. The main methods of sample augment include geometric transformation, color transformation, generative adversarial network generation, and manual modeling. Among them, geometric transformation and color transformation cannot effectively expand the diversity of sample features. Generative adversarial network can achieve good results for the augment of normal photovoltaic module, but hot-spot fault samples are diverse, and the use of generative adversarial network cannot change the characteristics of fault itself. In the theory of parallel vision, the manual modeling method can generate various types of hot-spot faults, which can effectively solve the problem of lacking enough training samples.

In view of the high modeling freedom of blender software, it is used to draw the structure, material, color, and background of photovoltaic module to create hot-spot fault models in this article. In order to improve the diversity of fault samples, different sizes and shapes of hot-spot areas are set at different positions. Meanwhile, image backgrounds such as roofs and mountains are used to simulate the actual scenes, which can ensure that the generated artificial samples conform to the physical characteristics and laws of real hot-spot faults. In the end, a total of 1000 artificial fault samples are obtained. The comparison between some artificial samples and real samples in different scenarios is shown in Fig. 3. It can be seen that the artificial samples have a high similarity to real samples, which can better assist real samples training.

B. Hot-Spot Fault Detection Network Based on Multiscale Feature Fusion

To improve the detection performance of multiscale hot-spot faults in complex background, a hot-spot fault detection algorithm based on multiscale feature fusion is proposed, abbreviated as MSFFNet. The structure is shown in Fig. 4.

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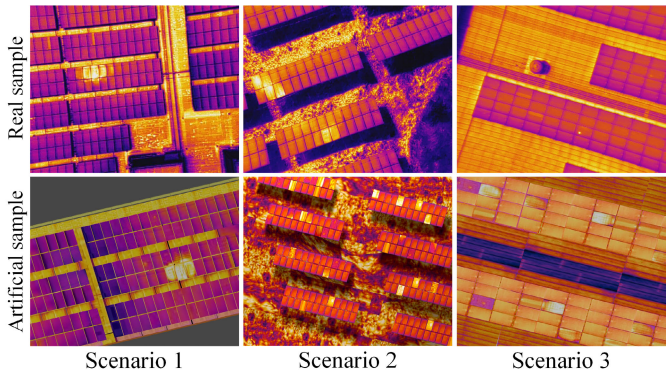


Fig. 3. Comparison between artificial and real samples in different scenarios.

1) Backbone Network Based on SCConv: To meet the needs of detection accuracy and speed, the SCConv [20] is introduced into the backbone network to replace conventional convolution. It solves the problem of feature redundancy from feature map space and channel dimension to realize the structural compression optimization of the model. The SCConv module is divided into two parts: the spatial reconstruction unit (SRU) and channel reconstruction unit (CRU). Moreover, it effectively reduces the parameters and computational costs required for feature extraction, making the model more compact, and also improving the detection accuracy.

For the input feature X, the SRU module is first used to reduce spatial redundancy features to obtain the spatial refined feature X^w. Then, the CRU module is further used to reduce channel redundancy features to obtain the channel refined feature Y. The above two reconstruction units are placed sequentially to form the SCConv module, as shown in Fig. 5.

The SRU module suppresses spatial redundancy by separating and reconstructing two parts to obtain the spatial thinning feature X^w, as shown in Fig. 6.

The SRU module first generates the informative feature map W_1 and sparse redundant map W_2 through channel feature separation. Then, W_1 and W_2 are dot-multiplied element by element with the input intermediate feature map X to obtain important feature X_1^w and redundant feature X_2^w. Split-cross-reconstruction is applied to X_1^w and X_2^w to generate the features X^{w1} and X^{w2}. Finally, X^{w1} and X^{w2} are concatenated to generate feature map X^w with spatial details. The calculation of the whole process is as follows:

$$\begin{cases} X_1^w = W_1 \otimes X \\ X_2^w = W_2 \otimes X \\ X_1^w = \left[X_{11}^w; X_{12}^w\right] \\ X_2^w = \left[X_{21}^w; X_{22}^w\right] \\ X_{11}^w \oplus X_{22}^w = X^{w1} \\ X_{21}^w \oplus X_{12}^w = X^{w2} \\ X^{w1} \coprod X^{w2} = X^w \end{cases} \tag{1}$$

where $\otimes$ represents the multiplication-by-element operation, $[\cdot, \cdot]$ represents the channel split operation, $\oplus$ represents the element-by-element addition operation, and $\coprod$ represents the channel concatenation operation.

Due to the redundancy of feature X^w in channel, the CRU module is used for split-transform-fuse strategy to reduce channel redundancy, as shown in Fig. 7.

In split stage, X^w is divided into aC and $(1-a)C$ along the channel dimension, and then 1×1 convolution is used to compress the channels of feature map to improve its computational efficiency. After split and compression operation, X^w is divided into the upper part X_{up} and lower part X_{low}. When X_{up} is input into the upper level transformation stage, the efficient and low-cost group wise convolution (GWC) and point wise convolution (PWC) are used to extract features to obtain the feature map Y_1, as shown in the following equation:

$$Y_1 = M^G X_{\text{up}} + M^P X_{\text{up}} \tag{2}$$

where M^G and M^P are learnable weights of GWC and PWC, respectively.

When X_{low} is input into the low-level transformation stage, a lightweight 1×1 PWC operation is used to generate feature maps with detailed information as a supplement to rich features. Then, the generated and reused features are concatenated to obtain feature map Y_2, as shown in the following equation:

$$Y_2 = M^P X_{\text{low}} \coprod X_{\text{low}}. \tag{3}$$

In fusion stage, SKNet [21] is used to adaptively fuse the features Y_1 and Y_2, which collects global spatial information with channel statistics through global average pooling (Pooling). The channel descriptor with global spatial information is as follows:

$$S_m = \text{Pooling}(Y_m) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} Y_c(i, j), m = 1, 2. \tag{4}$$

Then, the channel descriptors S_1 and S_2 of the upper and lower branches are stacked together, and important feature vectors β_1 and β_2 are generated by channel attention operation, as shown in in the following equation:

$$\beta_1 = \frac{e^{S_1}}{e^{S_1} + e^{S_2}}, \beta_2 = \frac{e^{S_2}}{e^{S_1} + e^{S_2}}, \beta_1 + \beta_2 = 1. \tag{5}$$

Finally, Y_1 and Y_2 are combined to obtain the channel refined feature Y, as shown in the following equation:

$$Y = \beta_1 Y_1 + \beta_2 Y_2 \tag{6}$$

2) Neck Network Based on RepG-MSFF: To improve the capacity of feature aggregation between different scales, the traditional FPN+PAN structure transmits strong semantic information from bottom to top while also transmitting strong localization information from top to bottom [22]. However, different scales feature maps are directly aggregated after size adjustment in this structure, which cannot fully utilize the multiscale information at the input end, thus affecting detection accuracy. To solve this issue, a RepG-MSFF feature pyramid structure is designed in this article. At the same time, a multiscale feature fusion (MSFF) module is also designed, which assigns weights to multiscale features by utilizing learnable weight parameters, and thus enhances

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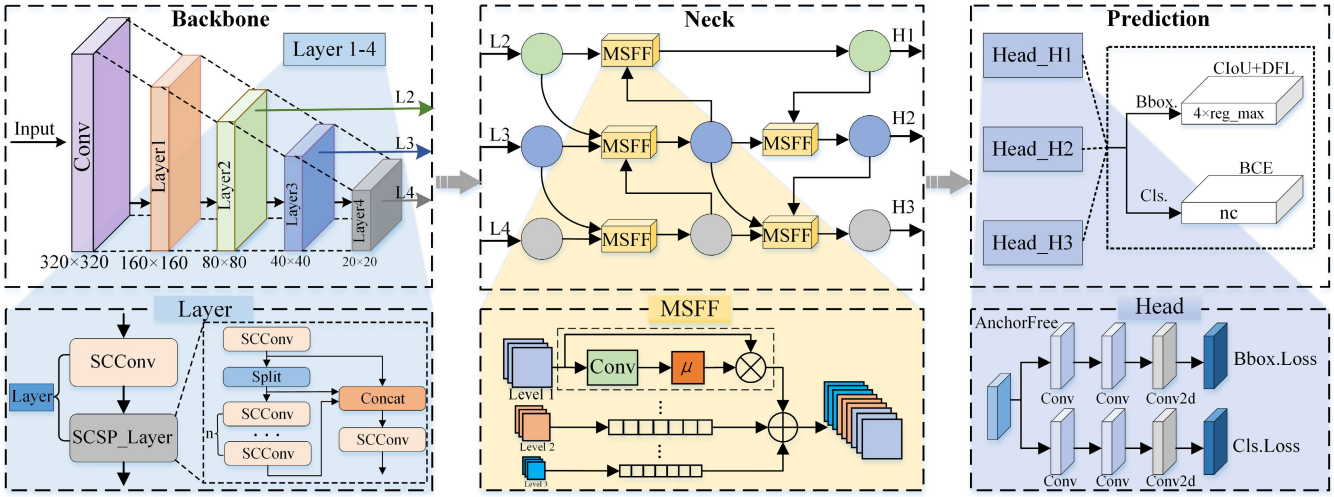


Fig. 4. Block diagram of MSFFNet.

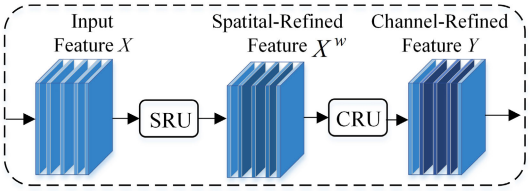


Fig. 5. SCConv module structure diagram.

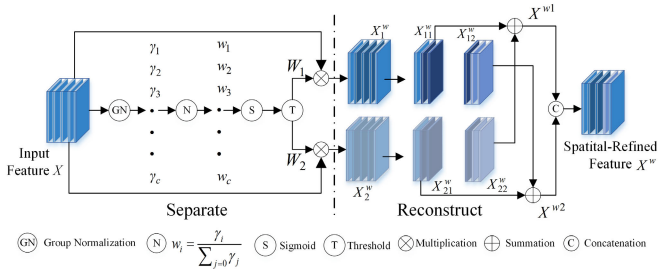


Fig. 6. Structure diagram of SRU.

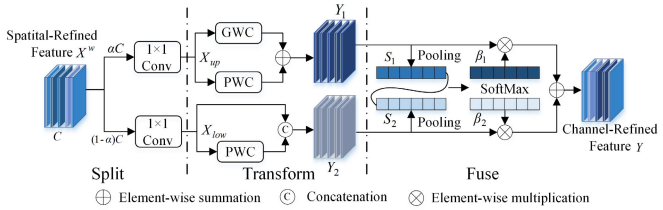


Fig. 7. Structure diagram of channel reconfiguration unit.

the detection performance for multiscale hot-spot fault targets. In the RepG-MSFF structure, cross-scale connection is adopted, which can flexibly control the expression ability of high-level and low-level features. Meanwhile, the designed MSFF module effectively increases the interactive fusion of features at different scales, providing a more comprehensive and rich feature representation, thereby adding more spatial details on the basis of high-level semantic features and enhancing the detection model's perception ability of the hot-spot targets. The comparison of feature pyramid structures between FPN+PAN and RepG-MSFF is shown in Fig. 8.

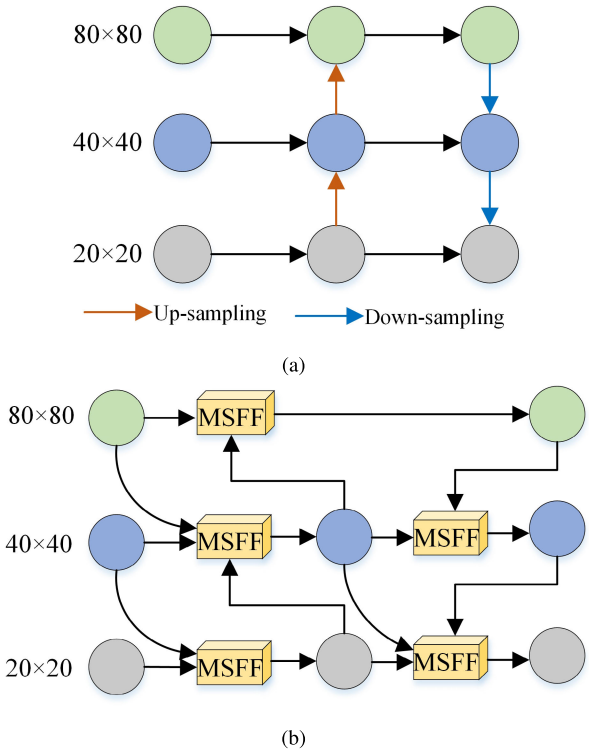


Fig. 8. Comparison of feature pyramid structures between FPN+PAN and RepG-MSFF. (a) Structure of FPN+PAN. (b) Structure of RepG-MSFF.

The designed RepG-MSFF feature pyramid structure can effectively fuse different scale features from adjacent upper and lower layers as well as the same level, which can enhance the capability of feature reuse and feature interaction.

The FPN-PAN structure directly adds the input features from different scales after scaling adjustment, which is easy to cause the problem of insufficient feature fusion. Therefore, this article designs an MSFF module, which can adaptively learn the weight parameters of input feature. The model structure is shown in Fig. 9.

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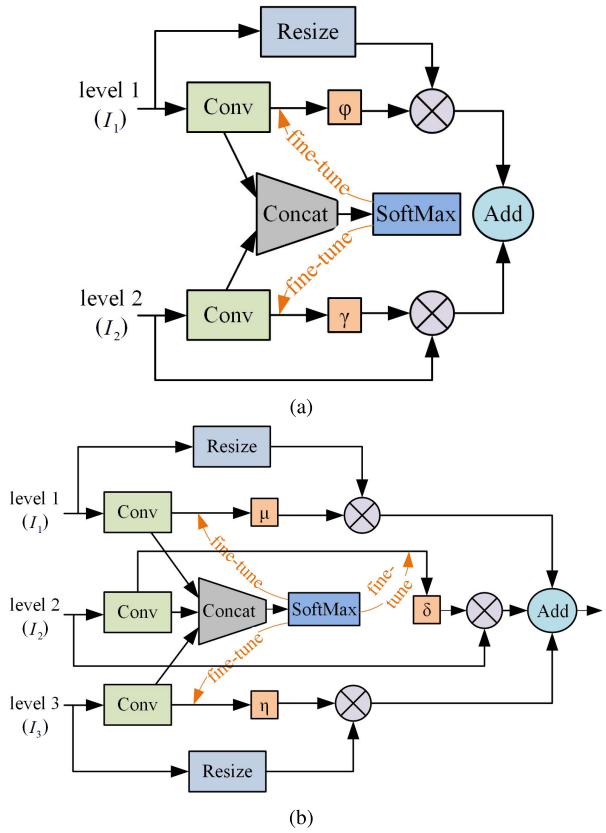


Fig. 9. MSFF module structure. (a) Two-scale feature fusion module. (b) Three-scale feature fusion module.

The input features of level i are denoted as I_i, where $i \in [1, 2, 3]$. First, scale adjustment is applied to features of different scales. In the up-sampling process, 1×1 convolution is used to adjust the channels, and nearest neighbor interpolation is introduced to improve the resolution. In the down-sampling process, a 1/2-scale down-sampling is completed by a 3×3-convolution with a step size of 2, and a 1/4-scale down-sampling is completed by adding a max pooling layer with a step size of 2 before the two-step convolution. Then, the weight parameters of each input features are predicted using a 1×1-convolution, and the algorithm convergence is accelerated while maintaining the relative relationship between weight parameters. The SoftMax function is used to normalize each predicted weight. Finally, after the scale adjustment, the input features are weighted to obtain the fused output features, as shown in the following equation:

$$\mathrm{MSFF}(I_1, I_2) = \varphi \cdot I_1^{1\to l} + \gamma \cdot I_2^{2\to l} \tag{7}$$

$$\mathrm{MSFF}(I_1, I_2, I_3) = \psi \cdot I_1^{1\to l} + \delta \cdot I_2^{2\to l} + \eta \cdot I_3^{3\to l} \tag{8}$$

where $I_i^{i\to l}$ represents the feature information after adjusting level i to level l, and φ, γ, ψ, δ, and η represent the learnable weight parameters of each multiscale feature after normalization by SoftMax function. The values are as follows:

$$\varphi + \gamma = 1, \varphi = \frac{e^{\lambda_\varphi}}{e^{\lambda_\varphi} + e^{\lambda_\gamma}} \tag{9}$$

$$\psi + \delta + \eta = 1, \psi = \frac{e^{\lambda_\psi}}{e^{\lambda_\psi} + e^{\lambda_\delta} + e^{\lambda_\eta}} \tag{10}$$

where λ_φ, λ_γ, λ_ψ, λ_δ, and λ_η represent the predicted weight parameters obtained by using 1×1 convolution.

3) Adaptive Threshold Focus Loss Function: Due to the complex geographical environment where photovoltaic modules are located, the hot-spot fault targets are easily submerged in backgrounds, which lead to an extreme imbalance between hot-spot targets and backgrounds. Therefore, in order to further accurately detect hot-spot targets, this article introduces an adaptive threshold focus loss (ATFL) function [23]. First, the hot-spot targets are decoupled from the background by setting the threshold value. The TFL function is as follows:

$$\mathrm{TFL} = \begin{cases} -(\lambda - P_t)^\eta \log(P_t), & P_t \le 0.5 \\ -(1 - P_t)^\gamma \log(P_t), & P_t > 0.5 \end{cases} \tag{11}$$

$$P_t = \begin{cases} P, & \text{if } y = 1 \\ 1 - P, & \text{others} \end{cases} \tag{12}$$

where η, γ, and λ are the hyperparameters, P_t represents the current average prediction probability value, P represents the prediction probability, and y represents the value of real label.

Second, for background information, the loss value should decrease with the increase of P_t, further reducing the loss generated by background. The predicted probability values can be expressed as follows:

$$\hat{P}_c = 0.05 \times \frac{1}{t-1} \sum_{i=0}^{t-1} \bar{P}_i + 0.95 \times P_t \tag{13}$$

where t represents the iteration epoch, $\hat{P}_c$ represents the predicted value of next epoch, and $\bar{P}_i$ represents the average prediction probability of each trained epoch.

Then, by intensifying the loss associated with hot-spot targets and reducing the loss associated with background, the model can allocate more attention to hot-spot features. According to the Shannon's information theory, the greater the probability value of an event, the smaller the amount of information it brings. Conversely, the greater the amount of information. Thus, γ and λ can be expressed as follows:

$$\gamma = -\ln(\hat{P}_c) \tag{14}$$

$$\eta = -\ln(P_t). \tag{15}$$

Finally, by incorporating (14) and (15) into (11), the expression of the ATFL can be obtained as follows:

$$\mathrm{ATFL} = \begin{cases} -(\lambda - P_t)^{-\ln(P_t)} \log(P_t), & P_t \le 0.5 \\ -(1 - P_t)^{-\ln(\hat{P}_c)} \log(P_t), & P_t > 0.5. \end{cases} \tag{16}$$

By introducing the ATFL function, the hot-spot targets can be separated from the complex backgrounds, and the loss weights can be adjusted by the adaptive mechanism, which improves the focus of detection model on hot-spot targets.

III. Experimental Results and Analysis

The configurations of the experimental platform are shown in Table II.

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TABLE II
HARDWARE AND SOFTWARE PLATFORM CONFIGURATIONS

Configuration	Version parameter
Operating system	Microsoft Windows 11
GPU	NVIDIA GeForce RTX 3070
CPU	AMD Ryzen 5 5600X 6-Core
CUDA	11.6
Deep lrarning framework	Pytorch

A. Experimental Dataset

The experimental dataset is composed of two parts: real samples and artificial samples, where a total of 1000 artificial hot-spot fault samples are generated by manual modeling. The real samples come from the UAV inspection provided by photovoltaic power station, which contains photovoltaic modules in three different scenarios. Among these real samples, 800 pictures are randomly selected for experiment, of which 600 pictures are taken as training samples and remaining 200 pictures are taken as test samples.

B. Network Model Training

During the training process, the momentum factor is set to 0.937, the bath size is set to 8 and SGD is taken as optimizer. The learning rate is set to 0.01 in the first 200 epochs and then decreased to 0.001 in the last 100 epochs. The weight decay regularization term is set to 0.005. Finally, after 300 epochs of model iterative training, the optimal weight is obtained.

C. Ablation Experiments of Artificial Sample Expansion

To objectively evaluate the advantages of proposed method, precision (P), recall (R), and average precision mean mAP@0.5 are selected, and the calculation formulas are as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \times 100\% \tag{17}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \times 100\% \tag{18}$$

$$\mathrm{mAP@0.5} = \int_0^1 P(R)\mathrm{dR} \tag{19}$$

where TP indicates correctly detected hot-spot faults, FP indicates incorrectly detected hot-spot faults, and FN indicates undetected hot-spot faults.

1) Ablation Experiment of Mixed Sample: To verify the influence of expanded number of artificial samples on detection results, this article adds different numbers of artificial samples to form mixed sample dataset for training on the basis of 600 real samples. During the testing process, the test dataset is kept unchanged, and artificial samples are only used to expand the training dataset. The detection results of different numbers of artificial samples are shown in Table III.

Since the artificial samples have similar characteristics to real samples, they can help the network better learn the characteristics of hot-spot faults. Therefore, expanding artificial samples can improve the detection accuracy and robustness.

TABLE III
INFLUENCE OF DIFFERENT NUMBER OF MANUAL SAMPLES ON TEST RESULTS

number of manual samples	mAP@0.5(%)	R(%)	P(%)
0	81.9	77.2	83.8
200	82.3	77.5	84.3
400	82.9	76.6	84.5
600	83.6	78.1	**86.2**
800	**85.8**	**78.4**	85.3
1000	83.8	77.3	84.6
1200	82.4	76.2	83.5
1400	82.1	75.9	82.8

TABLE IV
INFLUENCE OF TRANSFER LEARNING ON DETECTION EFFECT

Migration model	mAP@0.5(%)	R(%)	P(%)
Non-use	81.9	77.2	83.8
COCO transfer learning	82.2	74.3	80.6
VOC transfer learning	81.6	75.1	80.3
Artificial sample transfer learning	**83.2**	**78.2**	**83.9**

However, the more artificial samples are not better. Due to the differences between artificial samples and real samples, when the expansion of artificial samples far exceeds that of real samples, the detection model will focus on artificial samples, while the detection ability of real samples will decrease. Therefore, according to the number of real samples, adding an appropriate number of artificial samples can better improve the detection accuracy. Through different experiments, it is found that when the proportion of artificial samples to real samples is 4:3, the detection accuracy is highest.

2) Ablation Experiment of Transfer Learning Training: To verify the effect of transfer learning based on artificial samples, several different pre-trained models are loaded, respectively, to carry out experiments, as shown in Table IV.

Compared with large-scale migration models such as COCO and VOC, transfer learning based on artificial samples is more conducive to improving the detection accuracy. The reason for this phenomenon is that the structural texture features of artificial samples are closer to real fault samples, which can better assist real samples in training. The comparison curve of loss function between artificial sample transfer learning and non-artificial sample transfer learning is shown in Fig. 10. After adopting the artificial sample transfer learning, the convergence speed is faster, the stable value is smaller, and the stability is better.

3) Mixed Sample and Transfer Learning Combined Experiment: To verify the validity of mixed sample and transfer learning strategies, these two strategies are jointly trained. In the experiments, the proportion of artificial samples to real samples is 4:3, and the transfer learning model is based on artificial samples.

From Table V, it can be seen that the application of mixed samples combined with transfer learning has a better

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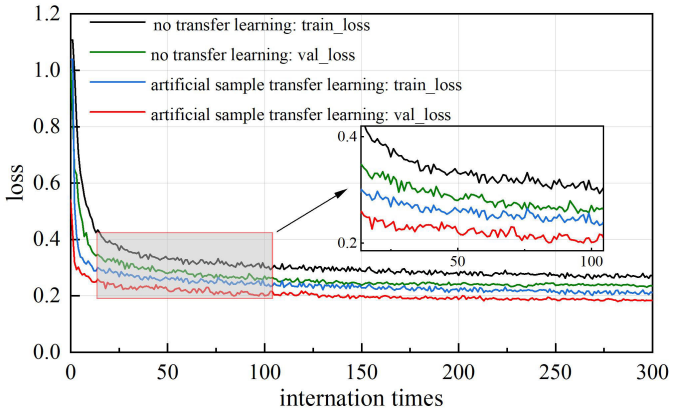


Fig. 10. Loss function comparison curve.

TABLE V
COMBINED EXPERIMENT OF THE TWO STRATEGIES

Mixed sample	Transfer learning	mAP@0.5(%)	R(%)	P(%)
		81.9	77.2	83.8
✓		85.8	78.4	85.3
	✓	83.2	78.2	83.9
✓	✓	**86.1**	**78.8**	**85.7**

TABLE VI
ABLATION EXPERIMENT OF DIFFERENT λ

λ	mAP@0.5(%)	R(%)	P(%)
Baseline	85.7	76.8	85.5
1.5	85.5	76.3	84.2
2.0	85.8	77.4	84.5
2.5	86.1	78.1	85.8
3.0	**86.5**	77.6	**86.6**
3.5	86.2	77.2	84.8
4.0	85.4	**78.3**	85.9

improvement effect, which effectively solves the problem of low detection accuracy caused by insufficient fault samples.

D. Ablation Experiments of ATFL Parameters Setting

To investigate the influence of λ value on ATFL function, different λ values are set for ablation experiments, and the experimental results are shown in Table VI.

From Table VI, it can be seen that when $\lambda = 3.0$, the optimal level is reached. Therefore, in the ATFL function, we set the hyperparameter λ to 3.0.

The ATFL decouples the hot-spot targets and background based on the set threshold, so the setting of threshold can affect detection performance. The detection results with different threshold settings are shown in Table VII. When the threshold is set to 0.5, the detection accuracy is highest.

E. Ablation Experiments of MSFFNet

To objectively evaluate the detection performance of the proposed method, a comparison is made between the

TABLE VII
ABLATION EXPERIMENT OF DIFFERENT THRESHOLD SETTING

Threshold setting	mAP@0.5(%)	R(%)	P(%)
Baseline	85.7	76.8	85.5
0.1	85.2	77.5	83.6
0.3	85.4	**77.8**	82.1
0.5	**86.3**	77.4	83.2
0.7	85.7	75.6	**84.5**
0.9	85.1	75.9	84.2

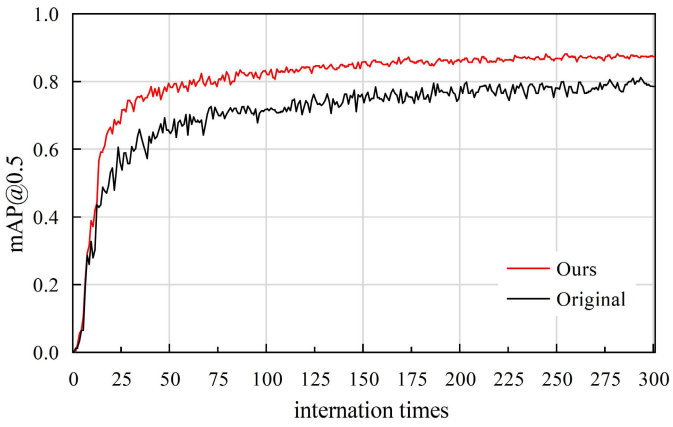


Fig. 11. mAP@0.5 comparison curve.

proposed algorithm and original YOLOv8 detection framework, as shown in Fig. 11.

The mAP@0.5 of the proposed method finally stabilizes at around 0.881, and the average detection accuracy is higher than original network, which shows that the proposed algorithm has a good detection performance for photovoltaic hot-spot faults.

In the detection algorithm MSFFNet, this article mainly proposes three improved strategies in backbone, neck, and loss function, which are as follows: backbone network based on SCConv, neck network based on RepG-MSFF, and ATFL function. To verify the advantages of these three parts, different improved strategies are added to conduct ablation experiments on the basis of sample expansion. The experimental results of three different scenarios are shown in Table VIII, where the optimal values are marked in bold.

Each improved strategy has a certain improvement in detection accuracy. Among them, sample expansion improves the sample diversity, and the detection accuracy of three scenarios is improved by 3.7%, 4.9%, and 3.1%, respectively. By introducing SCConv, the detection accuracy is improved while the parameters and computational cost required for feature extraction are reduced. The designed RepG-MSFF improves the interaction ability of multiscale features and further promotes the effective fusion of different sizes hot-spot targets. The introduced adaptive focus threshold loss function uses an adaptive mechanism to adjust the loss weight, which makes detection model pay more attention to hot-spot targets. Finally, by integrating various improved strategies, the detection accuracy of three scenarios can reach 87.9%, 88.6%,

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TABLE VIII
COMPARISON RESULTS OF ABLATION EXPERIMENTS

Sample expansion	Backbone SCConv	Neck RepGFPN	Loss ATFL	Scenario 1 mAP@0.5 (%)	Scenario 1 R (%)	Scenario 1 P (%)	Scenario 2 mAP@0.5 (%)	Scenario 2 R (%)	Scenario 2 P (%)	Scenario 3 mAP@0.5 (%)	Scenario 3 R (%)	Scenario 3 P (%)	Flops (G)
				81.5	77.7	84.5	82.4	75.6	84.3	83.1	77.4	82.1	8.4
✓				85.2	78.6	84.7	87.3	76.7	85.6	86.2	77.7	84.3	8.4
✓	✓			86.7	76.3	83.4	87.5	77.4	85.3	87.2	76.3	84.1	7.9
✓		✓		86.5	77.6	84.4	87.7	75.3	85.8	87.0	76.1	**84.6**	8.6
✓			✓	85.7	77.5	82.2	87.5	75.6	86.4	86.8	77.4	83.6	8.6
✓	✓	✓		85.8	78.1	84.7	87.9	76.8	**86.6**	87.3	**77.8**	83.8	8.1
✓	✓		✓	86.2	77.5	84.1	86.2	77.6	85.1	87.1	76.3	84.4	7.9
✓		✓	✓	87.3	78.7	85.1	88.4	**78.8**	85.5	87.5	76.5	84.2	8.6
✓	✓	✓	✓	**87.9**	**79.6**	**85.2**	**88.6**	78.1	85.7	**87.7**	77.5	84.1	**8.1**

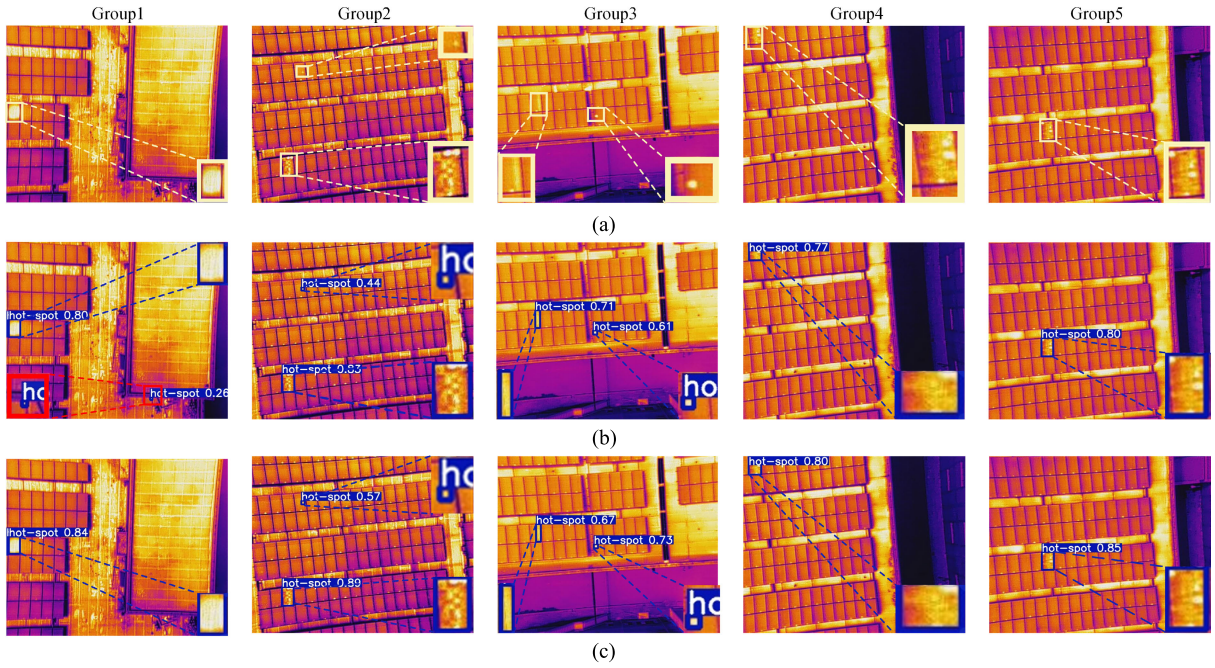


Fig. 12. Detection results of Scenario 1. (a) Fault image. (b) Original. (c) Ours.

and 87.7%, respectively, which is 6.4%, 6.2%, and 4.6% higher than original detection network.

To subjectively verify the advantages of proposed method, five groups of fault images in three different scenarios are selected for testing. The visualization results of original detection network and proposed algorithm are shown in Figs. 12–14, where the fault areas are marked with blue solid wire frame, the false and missed detection areas are marked with red solid wire frame, and the hot-spot faults are labeled as "hot-spot." From Figs. 12–14, the original algorithm has the phenomenon of false and missed detection, while the proposed algorithm has more accurate positioning and higher detection accuracy, fully proving that each improved strategy has significant effects on hot-spot fault detection.

F. Comparison With Advanced Methods

To further verify the advantages of proposed algorithm in hot-spot detection, nine detection algorithms including Dynamic-RCNN [24], VarifocalNet [25], TOOD [26], FasterNet [27], YOLOv5, YOLOv7 [28], YOLOv8, RT-DETR [29], and Gold-YOLO [30] are selected for comparison. Comparative experiment results are presented in Table IX. The mAP@0.5 values of Dynamic-RCNN algorithm in three different scenarios are 78.6%, 79.9%, and 72.4%, respectively. Its detection accuracy for hot-spot faults is obviously low. The VarifocalNet algorithm has the best precision (P) value in Scenario 3, which can reach 84.5%. However, it has a higher number of model parameters, occupying a significant amount of memory. The Flops of TOOD and YOLOv7 algorithms are

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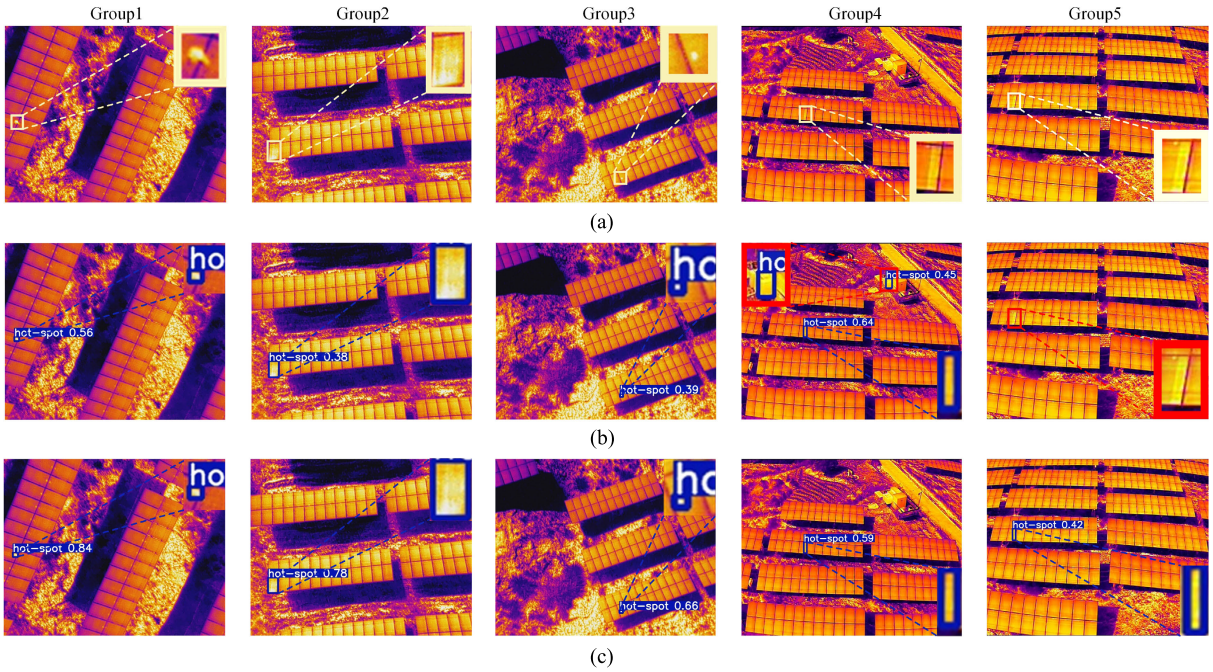


Fig. 13. Detection results of Scenario 2. (a) Fault image. (b) Original. (c) Ours.

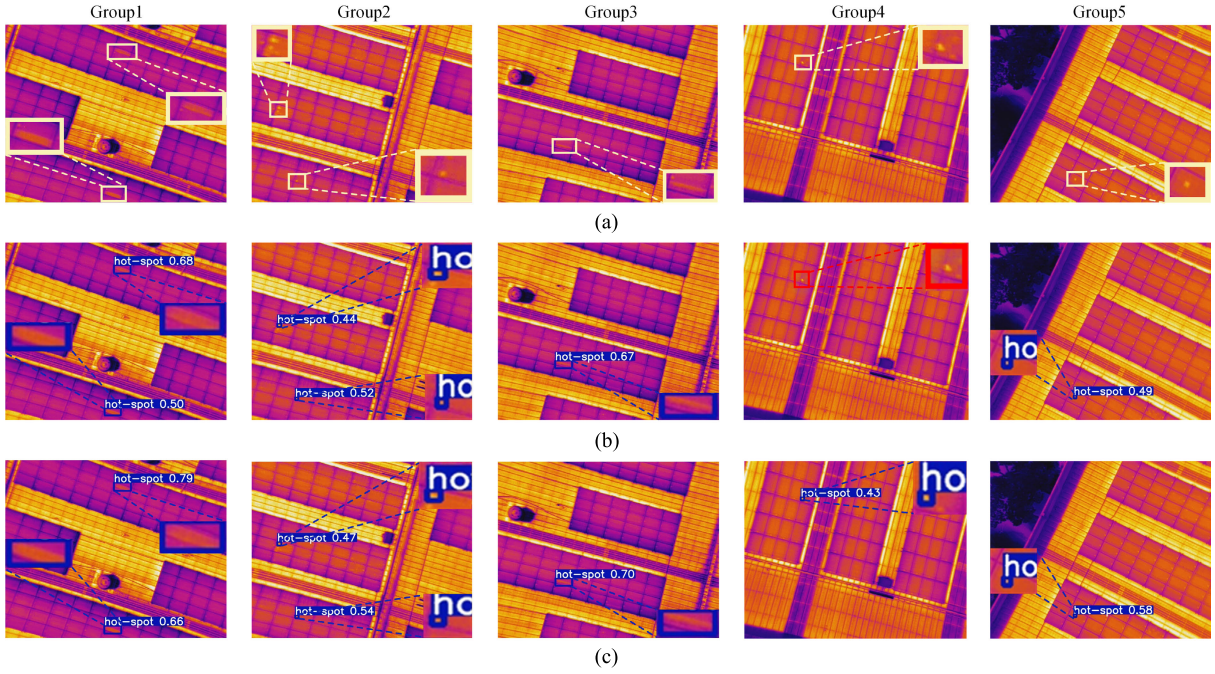


Fig. 14. Detection results of Scenario 3. (a) Fault image. (b) Original. (c) Ours.

72.16 and 105.1 G, respectively. The large number of Flops increases the complexity of detection model. The FasterNet and YOLOv8 algorithms have good performance in detection accuracy and speed, but compared to the proposed algorithm in this article, their detection effect for hot-spot faults still needs to be improved. The recall (R) value of YOLOv5 algorithm can reach 78.5% in Scenario 2, but its low detection accuracy for hot-spot faults is a notable drawback. Even though the RT-DETR and Gold-YOLO algorithms exhibit good detection accuracy, the large number of parameters and Flops increases the complexity of the detection model, and the detection efficiency is low. The mAP@0.5 values of the proposed algorithm in three different scenarios are 87.9%, 88.6%, and 87.7%, respectively, which are significantly higher than other comparison algorithms. At the same time, the proposed algorithm has the fast detection speed. Therefore, the proposed algorithm in this article is superior to other comparison algorithms.

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TABLE IX
DETECTION RESULTS OF DIFFERENT ALGORITHMS

Algorithm	Scenario 1				Scenario 2				Scenario 3				Parameters	Flops(G)
	mAP@0.5 (%)	R (%)	P (%)	Time (ms)	mAP@0.5 (%)	R (%)	P (%)	Time (ms)	mAP@0.5 (%)	R (%)	P (%)	Time (ms)		
Dynamic-RCNN	78.6	64.3	79.7	17.0	79.9	63.5	78.4	16.1	72.4	61.6	80.3	18.3	29653729	78.12
VarifocalNet	80.2	76.6	82.3	20.6	78.6	73.7	83.2	23.7	81.3	77.9	84.5	17.3	34057748	75.56
TOOD	81.3	79.5	84.6	20.2	82.5	77.4	81.6	15.7	75.4	73.6	81.2	22.5	32505856	72.16
FasterNet	83.8	78.1	83.4	16.6	84.8	5.7	83.4	14.6	82.8	75.1	83.5	16.7	57278649	25.9
YOLOv5	78.5	75.9	84.5	19.7	79.7	**78.5**	81.2	16.8	75.6	70.3	82.7	15.8	7015519	15.8
YOLOv7	73.7	68.5	75.3	18.4	74.6	70.8	73.9	19.2	72.5	69.4	74.7	18.6	37155845	105.1
YOLOv8	81.5	77.7	84.5	16.2	82.4	75.6	84.3	20.1	83.1	77.4	82.1	16.8	3486867	8.4
RT-DETR	84.2	77.8	83.8	15.2	85.4	74.6	81.5	15.3	83.2	**78.6**	83.2	16.3	38795276	115.7
Gold-YOLO	85.6	78.5	**86.3**	18.5	86.7	77.7	83.6	14.2	86.4	75.3	**84.6**	15.5	47323584	85.6
Ours	**87.9**	**79.6**	85.2	**15.0**	**88.6**	78.1	**85.7**	**13.8**	**87.7**	77.5	84.1	**15.1**	**3005843**	**8.1**

TABLE X
DETECTION RESULTS OF DIFFERENT INTERFERING FACTORS

Interference factors	Algorithm	Scenario 1			Scenario 2			Scenario 3		
		mAP@0.5(%)	R(%)	P(%)	mAP@0.5(%)	R(%)	P(%)	mAP@0.5(%)	R(%)	P(%)
Noise disturbance	Original	81.1	76.5	84.4	81.5	75.2	83.5	81.8	77.2	81.4
	Ours	**87.3**	**79.6**	**85.7**	**88.4**	**79.5**	**86.2**	**87.4**	**76.5**	**84.3**
Motion blur	Original	80.4	77.8	83.8	82.1	74.7	84.1	82.6	76.5	80.6
	Ours	**87.4**	**78.7**	**85.5**	**88.5**	**79.7**	**85.7**	**87.2**	**77.8**	**84.7**
Brightness and contrast	Original	80.5	76.4	83.8	81.7	74.4	83.8	82.5	76.1	80.9
	Ours	**87.7**	**79.2**	**85.3**	**88.2**	**78.4**	**85.2**	**86.8**	**77.4**	**84.1**

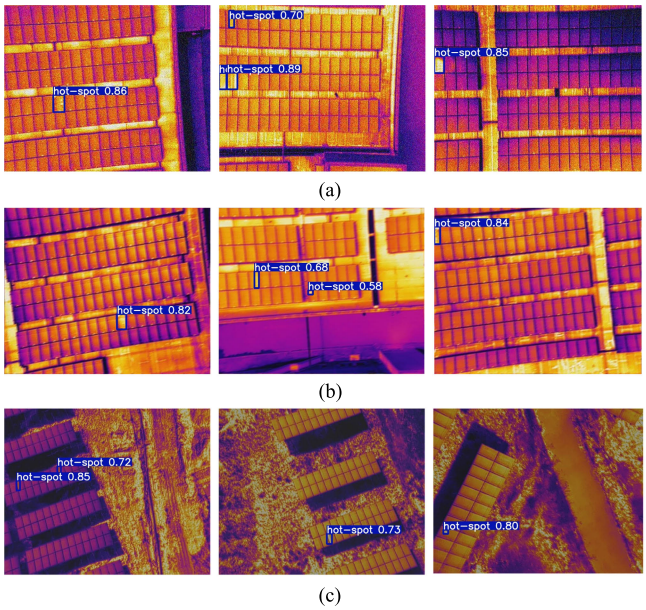


Fig. 15. Detection results under different interference factors. (a) Noise disturbance scenario. (b) Motion blur scenario. (c) Change in the brightness and contrast scenario.

G. Verification of Model Generalization Ability

To verify the generalization ability of detection model, the influence of real external factors is simulated by adding noise, motion blur, adjusting brightness, and contrast. The comparison of detection results between the original network and the proposed method is shown in Table X. After the addition of external factors, although the detection accuracy has decreased, the proposed method still has obvious advantages compared with the original network. The subjective detection results are shown in Fig. 15, from which it can be seen that the proposed method can still accurately detect hot-spot faults, which proves that the proposed model has a strong generalization ability.

IV. CONCLUSION

In this study, a novel hot-spot fault detection method is proposed, which takes parallel vision as the framework, and effectively improves the hot-spot detection performance under small samples through sample expansion, transfer learning, and MSFFNet design. The main conclusions are as follows.

1) A pre-training model based on artificial samples is established and transferred to the training of mixed samples, which not only improves the detection performance and robustness, but also effectively solves the problem of less common features in cross-domain transfer learning.
2) The designed RepG-MSFF feature pyramid structure can realize the effective fusion of different scales hot-spot features, effectively enhancing feature reuse and interaction capabilities.

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3) The introduced SCConv solves the problem of feature redundancy from the space and channel dimension, thereby balancing the needs of detection accuracy and speed.
4) The improved ATFL function separates hot-spot targets from complex backgrounds, which uses an adaptive mechanism to adjust loss weight to improve the focus of detection model on hot-spot targets.

By conducting comparative experiments on three different scenarios datasets provided by photovoltaic power station, the detection accuracy can reach 87.9%, 88.6%, and 87.7%, respectively, which is higher than other eight classical detection algorithms. However, in this study, the infrared images primarily provide thermal distribution information and cannot directly reflect the other conditions of photovoltaic modules, which results in certain limitations when identifying the causes of faults. Therefore, future work will focus on combining infrared images with other data types, such as visible images, to achieve a more comprehensive assessment of the actual conditions of photovoltaic modules through multisource information fusion, thereby improving the accuracy of hot-spot faults detection.

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Shuai Hao was born in Cangzhou, Hebei, in 1986. He received the B.S. degree from Xi'an University of Science and Technology, Xi'an, China, in 2010, and the Ph.D. degree from the School of Automation, Northwestern Polytechnical University, Xi'an, in 2014.

He is currently an Associate Professor with Xi'an University of Science and Technology. His research interests include pattern recognition, artificial intelligence, and digital image processing technology.

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Jiahao Li was born in Xi'an, Shaanxi, in 1999. He received the B.S. degree from the School of Electrical and Control Engineering, Xi'an University of Science and Technology, Xi'an, China, in 2022, where he is currently pursuing the M.S. degree.

His research interests include photovoltaic panel defect detection based on machine vision.



Xu Ma was born in Xi'an, Shaanxi, in 1985. She received the Ph.D. degree from the School of Automation, Northwestern Polytechnical University, Xi'an, China, in 2017.

She is currently a Lecturer with the School of electrical and control engineering, Xi'an University of Science and Technology, Xi'an. Her research interests include vision navigation and digital image processing.



Siya Sun was born in 1986. She received the M.S. degree in power system and automation from School of Electrical and Control Engineering, Xi'an University of Science and Technology, Xi'an, China, in 2011, where she is currently pursuing the Ph.D. degree in mechanical engineering.

Her research interests include cooperative control, strategic planning, and precise localization.



Zhuo Tian was born in Baoji, Shaanxi, in 1998. He received the B.S. degree from the School of Electrical and Control Engineering, Xi'an University of Science and Technology, Xi'an, China, in 2020, where he is currently pursuing the M.S. degree.

His research interest includes electrical equipment fault detection based on deep learning.



Tianqi Li was born in Luoyang, Henan, in 2000. He received the B.S. degree from Zhongyuan University of Technology, Zhengzhou, China, in 2023. He is currently pursuing the M.S. degree in control science and engineering with the School of Electrical and Control Engineering, Xi'an University of Science and Technology, Xi'an, China.

His research interests include image fusion and target detection.



Yifeng Hou was born in Yulin, Shaanxi, in 2004. He is currently pursuing the B.S. degree with the School of Electrical and Control Engineering, Xi'an University of Science and Technology, Xi'an, China.

His research interests include 3-D modeling and animation simulation.

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